

ZACH PARK

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EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering - Research

Pittsburgh, PA

May 2028

- Current Coursework: Machine Learning and AI for Engineers, Trustworthy AI

The Cooper Union for the Advancement of Science and Art

GPA: 3.86/4.0 (Summa Cum Laude) | New York, NY

Bachelor of Engineering in Mechanical Engineering, Minor in Bioengineering

May 2026

- Related Coursework: Manufacturing Engineering, Design & Prototyping, Mechatronics, Signal Processing

PROFESSIONAL EXPERIENCE

Product Design Intern, HUROTICS Inc.

Seoul, South Korea | Jun. 2026 – Aug. 2026

- Designed company's first gait test robot in SolidWorks and developed LabVIEW-based actuator control architecture for automated exosuit validation, resulting in company-funded \$10K production build
- Improved existing gait-labeling accuracy by 13% through ToF-IMU sensing system spanning ML-based classification, sensor hardware, Teensy firmware, MATLAB signal processing, and feature extraction
- Detected upcoming terrain 34% earlier than previous sensing system by developing Python LiDAR-camera-IMU perception pipeline using multimodal sensor fusion, optical-flow tracking, depth-profile analysis, and regression-based prediction
- Collaborated with apparel and embedded teams to design self-locking anchors, actuator docking, and thermal-management hardware for next-generation exosuit, with multiple components selected for final production design

Mechanical Design Intern, Mount Sinai Hospital BioDesign

New York, NY | Aug. 2025 – May 2026

- Reduced fabrication time by 75% and material cost by 40% by prototyping reusable silicone intracerebral hemorrhage (ICH) brain-and-skull phantom in Fusion 360 for neurosurgical training and device validation
- Created patient-specific chronic subdural hematoma (CSDH) testbed from CT anatomy with <1 mm cranial deviation, replaceable dura, silicone brain, and configurable burr-hole access for medical device testing
- Increased endoscopic field of view by 17% through 4 CAD design iterations with neurosurgeons and industry partners

PROJECT EXPERIENCE

Active Armrest Robot for Microsurgery Clinicians, Cooper Union

New York, NY | Aug. 2025 – May 2026

- Led 3-engineer interdisciplinary team with Mount Sinai clinicians to develop stool-mounted active armrest, integrating C++ control, lead-screw actuation, ToF sensing, and machined hardware for hands-free height adjustment
- Developed and CNC-machined 6061-aluminum two-link arm with GD&T and tolerance-controlled fits for smooth, low-friction passive motion, validating 7 kg design load through FEA with factor of safety >7
- Lowered trapezius EMG RMS by 30%, drawing deviation by 23%, and forward-head offset by 11% through sEMG signal processing, DWPose computer-vision posture tracking, and precision-task validation

Wearable Sensor for Dialysis Access Monitoring, Cooper Union

New York, NY | Jan. 2025 – May 2026

- Partnered with Mount Sinai nephrologists to engineer dual-axis piezoelectric wearable for arteriovenous fistula (AVF) vibration monitoring, combining orthogonal sensing, differential amplification, and LabVIEW-based signal processing
- Fabricated <\$200 vascular test rig with custom pulsator and compliance chamber, replacing \$3,000+ commercial pump, supporting 30+ validation trials, and contributing to 2nd Place and \$15K award in 2025 NIH DEBUT Challenge

Deep Learning for Electromagnetic Forming, Technische Universität Dresden

Dresden, Germany | May 2024 – Aug. 2024

- Integrated LS-DYNA electromagnetic-forming simulations with Python/PyTorch deep-learning pipeline, training Temporal Fusion Transformer to predict sheet-metal forming depth from material, process, and measured-current data
- Cut predictive coefficient of variation by up to 66.7% by guiding uncertainty-driven DoE on 210-sample dataset, using parameter-specific Monte Carlo dropout to prioritize 29 additional physical tests with Fraunhofer IWU

SKILLS

Languages: Python, C/C++, MATLAB

Tools: SolidWorks, Fusion 360, ANSYS, LS-DYNA, LabVIEW/NI DAQ, ROS 2, PyTorch, OpenCV

Methods: Rapid Prototyping, GD&T, Tolerance Analysis, DFM/DFA, FEA, Sensor Fusion